One Edge, Unbounded Instability:

Hilbert Decomposition Signed Measures in Multiparameter Persistence

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Abstract

We prove that Hilbert decomposition signed measures, a computable summary of multiparameter persistent homology, are not uniformly stable over all finite filtrations once there are at least three parameters. For every $n \geq 3$ and $m \geq 1$, we construct two filtrations of the same finite graph that differ only in the grade of one edge, yet

$$\|f_m - g_m\|_1 = 1, \qquad \|\mu_{H_0(f_m)} - \mu_{H_0(g_m)}\|_{\mathrm{KR},1} = 2m.$$

The output distance is therefore unbounded at fixed input distance one. This settles an open question of Loiseaux et al. [1] and gives a sharp dimensional cutoff: the dimension-only estimate holds for one and two parameters, but no finite dimension-only constant exists for $n \geq 3$. Rescaling puts every grade in the unit cube with input distance $1/m$ and output distance two, ruling out a graph-independent modulus of continuity. The four-vertex diamond graph is the smallest violation; after subdivision, the same mechanism occurs in lower-star filtrations. The corresponding module-level bound also fails. The counterexample family and its main quantitative consequences are formalized and kernel-checked in Lean.

1 Introduction

Multiparameter persistent homology indexes a filtration by vectors in $\mathbb{R}^n$. The resulting modules rarely admit barcodes, so topological data analysis uses computable summaries such as Hilbert functions, multigraded Betti numbers, rank invariants, and signed decompositions [4, 6]. A Hilbert decomposition encodes the Hilbert function as a finite signed point measure. These measures support stable vectorizations for machine learning [1] and are implemented in `multipers` [8].

Loiseaux et al. proved that, for a finite simplicial complex S and monotone maps $f, g : S \to \mathbb{R}^n$,

$$\|\mu_{H_i(f)} - \mu_{H_i(g)}\|_{\mathrm{KR},1} \leq n\|f - g\|_1 \qquad (n \in \{1, 2\}), \tag{1}$$

and explicitly left the extension to $n > 2$ open [1, Theorem 1(1)]. Oudot and Scoccola proved the corresponding one- and two-parameter module-level estimate [2, Theorem 1.5]. We show that both extensions fail in every parameter dimension $n \geq 3$. The failure already occurs in degree-zero persistent homology of graphs when the grade of a single edge changes.

Theorem 1.1 (Filtered-graph counterexample)**.** *Fix a coefficient field. For every $n \geq 3$ and every integer $m \geq 1$, there are a finite simple graph S_m and monotone filtrations $f_m, g_m : S_m \to \mathbb{R}^n$ such that*

$$\|f_m - g_m\|_1 = 1, \qquad \|\mu_{H_0(f_m)} - \mu_{H_0(g_m)}\|_{\mathrm{KR},1} = 2m.$$

Only one edge has different grades under f_m and g_m.

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The four-vertex case is displayed in section 3. The general construction and the proof of theorem 1.1, including the exact transport calculation and the extension from three to n parameters, are given in section 4.

Corollary 1.2 (Exact dimensional phase transition)**.** *A finite constant depending only on parameter dimension can bound the Hilbert signed-measure distance by filtration ℓ^1-displacement for all finite filtered complexes if and only if $n \leq 2$.*

Equation (1) gives the positive half of corollary 1.2, and theorem 1.1 gives the negative half. In fact, the failure is not specific to the proposed constant n: no finite constant depending only on dimension can work. After scalar rescaling, corollary 5.1 rules out even a graph-independent modulus of continuity on the unit cube. The graph in our construction grows with m. This is compatible with the fixed-complex Lipschitz theorem of Scoccola et al., whose constant may depend on the complex [5, Proposition E.3].

The smallest violation of the proposed three-parameter constant occurs at $m = 2$. Its graph is the diamond $K_4 \setminus \{\text{one edge}\}$, and its exact output cost is $4 > 3$. By proposition 3.1, no simple graph with fewer vertices or fewer edges can violate the constant.

The module-level bound fails as well. Let $M_m = H_0(f_m)$ and $N_m = H_0(g_m)$. Their cellular presentations have the same incidence matrix; one column label moves by ℓ^1-distance one. The path-metric presentation distance of Bjerkevik and Lesnick [3] then gives the following result.

Theorem 1.3 (Presentation-distance counterexample)**.** *For every $n \geq 3$ and $m \geq 1$, there are finitely presented n-parameter persistence modules M_m, N_m such that*

$$d^1_{\mathcal{I}}(M_m, N_m) \leq 1, \qquad d^{\mathrm{Hil}}_{W,1}(M_m, N_m) = 2m.$$

No finite dimension-only constant therefore bounds Hilbert signed 1-Wasserstein distance by $d^1_{\mathcal{I}}$ for fixed $n \geq 3$.

We prove theorem 1.3 in section 6. The unit-cube and lower-star variants, the minimality result, and the H_1/Euler calculation are proved in sections 3 and 5. Reproducibility information appears in section 7; Appendix A records the correspondence between the paper statements and their Lean formalization.

2 Hilbert signed measures and transport

Let K be a finite simplicial complex. A filtration $f : K \to \mathbb{R}^n$ is *monotone* when $f(\tau) \leq f(\sigma)$ coordinatewise whenever τ is a face of σ. Its sublevel complex at $x \in \mathbb{R}^n$ is

$$K^f_x = \{\sigma \in K : f(\sigma) \leq x\}.$$

Fixing a coefficient field, the persistent homology module has Hilbert function

$$h_{H_i(f)}(x) = \dim H_i(K^f_x).$$

For two filtrations of the same finite complex, we use the simplexwise ℓ^1 norm

$$\|f - g\|_1 = \sum_{\sigma \in K} \|f(\sigma) - g(\sigma)\|_1.$$

For the finitely presentable modules considered here, the *Hilbert decomposition signed measure* μ_M is the unique finite signed point measure satisfying

$$h_M(x) = \mu_M((-\infty, x]) \quad \text{for all } x \in \mathbb{R}^n, \qquad (-\infty, x] = \{y : y \leq x\}. \tag{2}$$

This is the Möbius inverse of the Hilbert function [1, Definition 4 and Appendix A.4]. We write $\mu_{H_i(f)}$ for the measure of $H_i(f)$.

If a finite signed atomic measure Δ has Jordan decomposition $\Delta = \Delta^+ - \Delta^-$ with equal positive and negative total mass, Loiseaux et al. define its Kantorovich-Rubinstein norm with ℓ^1 ground cost as follows [1, Definition 3]:

$$\|\Delta\|_{\mathrm{KR},1} = \inf_{\pi \in \Pi(\Delta^+, \Delta^-)} \int_{\mathbb{R}^n \times \mathbb{R}^n} \|x - y\|_1 \, d\pi(x, y). \tag{3}$$

All measures below are finite sums of unit atoms, so (3) is a finite transportation problem. The lower bound uses only the fact that distinct integer-lattice points have distance at least one. Explicit matchings give the upper bound.

The Euler decomposition signed measure [1, Definition 5] is

$$\mu_{\chi(f)} = \sum_i (-1)^i \mu_{H_i(f)}.$$

Unlike the degree-specific Hilbert measures, it satisfies the dimension-independent estimate

$$\|\mu_{\chi(f)} - \mu_{\chi(g)}\|_{\mathrm{KR},1} \leq \|f - g\|_1 \tag{4}$$

in every parameter dimension [1, Theorem 1(2)].

3 The four-vertex counterexample

The graph S_2 has vertices s, t, u_1, u_2, central edge $e = \{s, t\}$, and two two-edge paths $s - a_j - u_j - b_j - t$. Every vertex has grade $(0, 0, 0)$, the arm grades agree under f_2 and g_2, and only e moves:

	a_1	b_1	a_2	b_2	e
f_2	$(0,1,0)$	$(0,0,2)$	$(0,2,0)$	$(0,0,1)$	$(2,0,0)$
g_2	$(0,1,0)$	$(0,0,2)$	$(0,2,0)$	$(0,0,1)$	$(1,0,0)$.

The filtered diamond is shown in figure 1.

The signed-measure difference $\Delta_2 = \mu_{H_0(f_2)} - \mu_{H_0(g_2)}$ has positive atoms

$$(1,0,0),\ (1,2,2),\ (2,1,2),\ (2,2,1)$$

and negative atoms

$$(1,1,2),\ (1,2,1),\ (2,0,0),\ (2,2,2).$$

The supports are disjoint subsets of $\mathbb{Z}^3$, so every coupling costs at least four. Matching vertically between the layers $r = 1$ and $r = 2$ costs exactly four. Since only e moves, this proves $4 > 3 \cdot 1$.

Proposition 3.1 (Minimality of the diamond)**.** *Among simple graphs with arbitrary monotone three-parameter filtrations, a counterexample to the proposed constant* 3 *requires at least four vertices and at least five edges. Hence S_2 is simultaneously vertex-minimal and edge-minimal.*

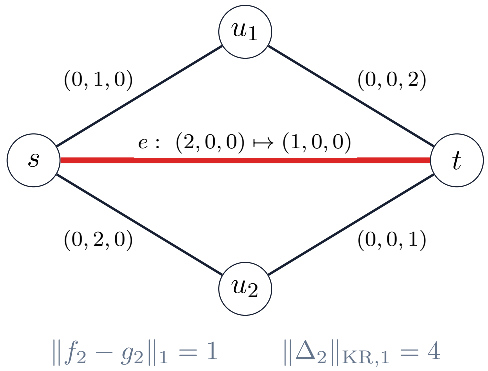


Figure 1: The diamond graph is the minimal counterexample: one edge moves by one, while the degree-zero Hilbert signed measure moves by four.

Proof. A simple graph with at most four edges has cycle rank at most one. If it is a forest, then $\mu_{H_1} = 0$ and $\mu_{H_0} = \mu_\chi$, so (4) gives the stronger constant 1.

If the graph is unicyclic, let C be its unique cycle. Its H_1 Hilbert function is the indicator of the upper orthant born at

$$b_f = \bigvee_{e \in C} f(e).$$

Monotonicity makes the vertex grades irrelevant once the cycle edges are present. Hence $\mu_{H_1(f)} = \delta_{b_f}$, and similarly for g. For each coordinate, the maximum operation is 1-Lipschitz, hence

$$\|b_f - b_g\|_1 \le \sum_{e \in C} \|f(e) - g(e)\|_1 \le \|f - g\|_1.$$

Using $\mu_{H_0} = \mu_\chi + \mu_{H_1}$ and the triangle inequality gives

$$\|\mu_{H_0(f)} - \mu_{H_0(g)}\|_{\mathrm{KR},1} \le 2\|f - g\|_1.$$

No graph with at most four edges can therefore violate the constant 3. A simple graph with at most three vertices has at most three edges, which also proves vertex-minimality. $\square$

4 The general family

4.1 Graph and filtrations

For $m \ge 1$, let S_m have vertices

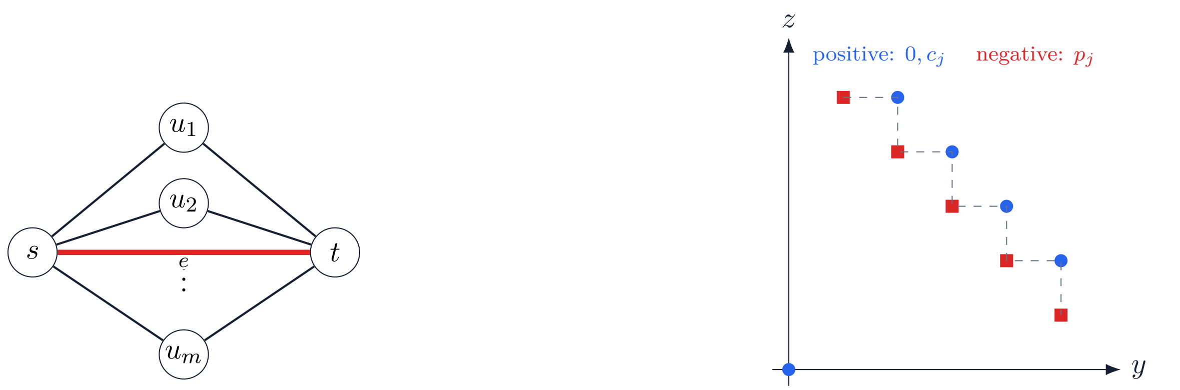
$$s, t, u_1, \dots, u_m$$

and edges

$$e = \{s, t\}, \qquad a_j = \{s, u_j\}, \qquad b_j = \{u_j, t\} \quad (1 \le j \le m).$$

There are no higher-dimensional simplices. Give every vertex grade $(0,0,0)$ and set

$$\begin{aligned} f_m(a_j) = g_m(a_j) &= (0, j, 0), \\ f_m(b_j) = g_m(b_j) &= (0, 0, m+1-j), \\ f_m(e) &= (2,0,0), \qquad\qquad g_m(e) = (1,0,0). \end{aligned}$$



S_m: m parallel two-edge paths

staircase measure ν_m (shown for $m = 5$)

Figure 2: Left: the graph family. Right: the two-dimensional staircase Möbius measure; the blue circle at the origin and the blue corner atoms alternate with the red path atoms.

All coordinates are nonnegative, so the filtrations are monotone. Only e moves, and therefore

$$\|f_m - g_m\|_1 = 1. \tag{5}$$

The construction and the staircase measure arising below are summarized in figure 2.

4.2 The component-count formula

Write a parameter as $x = (r, y, z)$. Outside the slab $1 \le r < 2$, the two sublevel graphs agree. Inside the slab, the g_m-sublevel graph contains e while the f_m-sublevel graph does not. Adding e merges two components exactly when no path $s - u_j - t$ is complete. The jth path is complete exactly when

$$y \ge j, \qquad z \ge m + 1 - j.$$

Define

$$q_m(y, z) = \begin{cases} 1, & y, z \ge 0 \text{ and no } j \text{ satisfies } y \ge j,\ z \ge m + 1 - j, \\ 0, & \text{otherwise.} \end{cases}$$

Since ordinary H_0 is the free vector space on connected components, over every coefficient field,

$$D_m(r, y, z) := \dim H_0(K^{f_m}_{(r,y,z)}) - \dim H_0(K^{g_m}_{(r,y,z)}) = \mathbf{1}_{[1,2)}(r) q_m(y, z). \tag{6}$$

4.3 Möbius inversion of the staircase

For $1 \le j \le m$, put

$$p_j = (j, m + 1 - j),$$

and, for $1 \le j < m$, put

$$c_j = p_j \vee p_{j+1} = (j + 1, m + 1 - j).$$

Define the signed point measure

$$\nu_m = \delta_{(0,0)} - \sum_{j=1}^{m} \delta_{p_j} + \sum_{j=1}^{m-1} \delta_{c_j}. \tag{7}$$

Lemma 4.1 (Staircase inversion)**.** *For every* $(y, z) \in \mathbb{R}^2$,

$$\nu_m((-\infty, y] \times (-\infty, z]) = q_m(y, z).$$

Proof. If $y < 0$ or $z < 0$, both sides vanish. Otherwise let

$$A(y, z) = \{j : p_j \leq (y, z)\}.$$

Because the first coordinates of the p_j increase while their second coordinates decrease, $A(y, z)$ is empty or a consecutive interval. If it has $a > 0$ elements, exactly $a - 1$ adjacent joins c_j lie below (y, z). The cumulative value of (7) is then $1 - a + (a - 1) = 0$. If $A(y, z)$ is empty, its value is 1. $\square$

Let $\eta = \delta_1 - \delta_2$ on $\mathbb{R}$. Its cumulative function is $\mathbf{1}_{[1,2)}$. Equations (2), (6), and lemma 4.1 give

$$\Delta_m := \mu_{H_0(f_m)} - \mu_{H_0(g_m)} = \eta \otimes \nu_m. \tag{8}$$

Write

$$\nu_m^+ = \delta_{(0,0)} + \sum_{j=1}^{m-1} \delta_{c_j}, \qquad \nu_m^- = \sum_{j=1}^{m} \delta_{p_j}.$$

Both have mass m, and no atom cancels in

$$\Delta_m^+ = \delta_1 \otimes \nu_m^+ + \delta_2 \otimes \nu_m^-, \tag{9}$$
$$\Delta_m^- = \delta_1 \otimes \nu_m^- + \delta_2 \otimes \nu_m^+. \tag{10}$$

The origin, the p_j, and the c_j are distinct in every potentially cancelling pair. Hence (9) and (10) are the Jordan parts, each with $2m$ unit atoms.

4.4 Exact transport and zero-padding

The two supports in (9) and (10) are disjoint subsets of $\mathbb{Z}^3$. Every distance from a source atom to a target atom is at least one, so every coupling costs at least $2m$. For the reverse bound, match $(1, a)$ to $(2, a)$ for every atom a of ν_m^+, and match $(2, b)$ to $(1, b)$ for every atom b of ν_m^-. Each of the $2m$ units moves by one, giving

$$\|\Delta_m\|_{\mathrm{KR},1} = 2m.$$

Equation (5) now proves theorem 1.1 for three parameters. For $n > 3$, append zero coordinates to every grade and atom. Monotonicity, sublevel components, and both ℓ^1 costs are unchanged.

5 Consequences and robustness

5.1 No graph-independent modulus on the unit cube

For $m \geq 2$, divide every grade by m. Denote the resulting filtrations by $\widetilde{f}_m, \widetilde{g}_m$. All grades lie in $[0, 1]^n$ and scalar dilation pushes forward the Hilbert signed measures. Since the ground metric is homogeneous,

$$\|\widetilde{f}_m - \widetilde{g}_m\|_1 = \frac{1}{m}, \qquad \|\mu_{H_0(\widetilde{f}_m)} - \mu_{H_0(\widetilde{g}_m)}\|_{\mathrm{KR},1} = 2. \tag{11}$$

Corollary 5.1. *For every fixed $n \geq 3$, the Hilbert signed-measure map on all finite n-parameter filtered graphs has no graph-independent modulus of uniform continuity, even with grades restricted to $[0,1]^n$. In particular, no bound*

$$\|\mu_{H_0(f)} - \mu_{H_0(g)}\|_{\mathrm{KR},1} \leq C\|f-g\|_1^{\alpha}$$

holds uniformly for any $C < \infty$ and $\alpha > 0$.

Proof. In (11), the input tends to zero while the output remains two. The Hölder conclusion follows by taking m large enough that $C/m^{\alpha} < 2$. □

Although the graph grows with m, it remains simple, planar, two-terminal series-parallel, and of treewidth two. Its size is

$$|V(S_m)| = m+2, \qquad |E(S_m)| = 2m+1, \qquad |S_m| = 3m+3$$

when vertices and edges are both counted as simplices. Along this family, any valid complex-dependent replacement constant must grow at least linearly: the ratio is $2(|V|-2) = \frac{2}{3}(|S_m|-3)$.

5.2 Lower-star filtrations

The freedom to grade edges independently is not essential.

Proposition 5.2 (Lower-star realization)**.** *For every $n \geq 3$ and $m \geq 1$, there are vertex functions on a subdivided finite graph, denoted v_m^f and v_m^g, whose induced lower-star filtrations f_m^{LS} and g_m^{LS} satisfy*

$$\|v_m^f - v_m^g\|_1 = 1, \qquad \|f_m^{\mathrm{LS}} - g_m^{\mathrm{LS}}\|_1 = 3, \qquad \|\mu_{H_0(f_m^{\mathrm{LS}})} - \mu_{H_0(g_m^{\mathrm{LS}})}\|_{\mathrm{KR},1} = 2m.$$

After division by m, the vertex-input distance is $1/m$, the induced simplexwise distance is $3/m$, and the output distance is 2.

Proof. Subdivide every edge of S_m once. Give each original vertex grade zero and give the midpoint of an old edge its old edge grade. In the induced lower-star filtration, the midpoint and both incident subdivided edges enter at exactly the old edge grade. Replacing an active old edge by this active two-edge path preserves connected components at every parameter, so the H_0 Hilbert functions and signed measures are unchanged.

Only the midpoint of the central edge has different vertex grades. Its own grade and the grades of its two incident subdivided edges each move by one, giving total filtration displacement three. Rescaling gives the last claim. □

5.3 Same-family H_1 and Euler cancellation

The dimension-independent Euler bound (4) may seem surprising next to theorem 1.1. The same graph family explains the difference. Set

$$\sigma_m = \sum_{j=1}^{m} \delta_{p_j} - \sum_{j=1}^{m-1} \delta_{c_j}, \qquad \nu_m = \delta_{(0,0)} - \sigma_m.$$

Let $\Delta_i = \mu_{H_i(f_m)} - \mu_{H_i(g_m)}$ and $\Delta_\chi = \Delta_0 - \Delta_1$.

Proposition 5.3 (Atomwise Euler cancellation)**.** *For the family in theorem 1.1,*

$$\Delta_0 = \eta \otimes (\delta_{(0,0)} - \sigma_m), \qquad \Delta_1 = -\eta \otimes \sigma_m, \qquad \Delta_\chi = \eta \otimes \delta_{(0,0)}.$$

Moreover,

$$\|\Delta_0\|_{\mathrm{KR},1} = 2m, \qquad \|\Delta_1\|_{\mathrm{KR},1} = 2m-1, \qquad \|\Delta_\chi\|_{\mathrm{KR},1} = 1.$$

Proof. For a sublevel graph,

$$\beta_1 = |E| - |V| + \beta_0.$$

Inside the slab, g_m has exactly one more active edge than f_m whenever the common vertex grade is active. Hence the H_1 Hilbert-function difference is the H_0 difference with the upper-orthant indicator born at $(1,0,0)$ and dying at $(2,0,0)$ removed. Möbius inversion gives the three displayed identities.

The H_1 Jordan parts have $m + (m-1) = 2m-1$ atoms each: corner atoms on one layer and path atoms on the other. Their supports are disjoint integer lattices and the vertical matching has unit cost per atom. The Euler difference has just the two atoms $(1,0,0)$ and $(2,0,0)$. □

In the alternating Euler sum, all staircase atoms cancel and only the moved simplex remains.

6 The module-level presentation distance

For a filtered graph, the cellular sequence

$$C_1(f_m) \xrightarrow{\partial_1} C_0(f_m) \longrightarrow H_0(f_m) \longrightarrow 0$$

is a finite labeled presentation. Choose the same orientations and simplex ordering for f_m and g_m. The resulting matrices are identical. Their row labels and arm-column labels agree, while the central column label moves from $(2,0,0)$ to $(1,0,0)$.

Bjerkevik and Lesnick define the preliminary 1-presentation distance between modules by minimizing the total ℓ^1 label displacement over compatible same-matrix presentations. Their presentation distance $d^1_{\mathcal{I}}$ is the path metric induced by this preliminary dissimilarity [3, Section 3.1 and Definition 3.2]; see also the complexity analysis of Bjerkevik and Botnan [7]. This compatible pair has cost one, so the one-step path gives

$$d^1_{\mathcal{I}}(H_0(f_m), H_0(g_m)) \le 1.$$

The pointwise cokernel of the incidence presentation is ordinary cellular H_0. Its Hilbert function is therefore the component-count function from section 4, whose signed-Wasserstein value is $2m$. This proves theorem 1.3.

After scaling every presentation label by $1/m$, we have

$$d^1_{\mathcal{I}}(M_m, N_m) \le \frac{1}{m}, \qquad d^{\mathrm{Hil}}_{W,1}(M_m, N_m) = 2.$$

The module invariant therefore has no dimension-only modulus of uniform continuity. The argument uses only the upper bound on $d^1_{\mathcal{I}}$; equality is neither needed nor claimed.

7 Artifact and reproducibility

The paper is accompanied by a kernel-checked Lean formalization and an independent finite computation. The accompanying repository contains the Lean source, build instructions, and pinned dependencies: Lean `v4.33.0-rc1` and Mathlib at the revision recorded in the Lake manifest. Running `lake build` from the repository root checks the complete formal development; the README gives prerequisites and additional audit commands. A NetworkX/NumPy/SciPy script independently reproduces the H_0 component counts, finite differences, and optimal assignments for $m = 1, \ldots, 15$. The computation is not used as a premise in Lean.

Use of AI-assisted tools. AI-assisted tools were used for literature triage, Lean formalization, and manuscript editing. The proofs were developed over the course of two months from conversations between the author and GPT-5.5 / 5.6 Sol. The author checked all mathematical statements and proof artifacts and takes responsibility for the content.

Competing interests. The author declares no competing interests.

Data and code availability. All constructions are finite and specified in the paper. Lean source, build metadata, and secondary verification scripts are available from `https://github.com/ZeterMordio/one-edge-instability` under the Apache-2.0 license. Appendix A gives the theorem-to-source map and the formal trust audit. The archived software release will identify its exact commit and DOI.

A Lean formalization certificate

The formal development uses Lean [9] and Mathlib [10]. The principal H_0 source file, `P1.lean`, imports only `Mathlib`. It belongs to the `OneEdgeInstability` library; the module-level result is proved separately in `P3.lean`. The development introduces the following definitions for this family:

- finite vertex, edge, and simplex types for S_m;
- real sublevel graphs and ordinary cellular $H_0 = C_0/\operatorname{im}\partial_1$;
- finite signed atomic measures, lower-orthant cumulative functions, and uniqueness of Möbius inversion;
- conversion to Mathlib signed measures and identification of their Jordan positive and negative parts;
- KR_1 as an infimum over Mathlib measures on $\mathbb{R}^n \times \mathbb{R}^n$, with measurable-set marginal equations and `lintegral` cost;
- labeled cellular presentations and the finite-path definition of $d^1_{\mathcal{I}}$;
- cellular $H_1 = \ker \partial_1$, field-independent rank, and the Euler-Poincaré bridge.

Paper statement	Lean declaration
Theorem 1.1	`P1.theorem1_H0_counterexample_family`
Failure of the constant-n bound	`P1.proposed_constant_n_P1_inequality_false`
No finite dimension-only constant	`P1.no_finite_dimension_only_P2_constant`
Diamond example $4 > 3$	`P1.n3_m2_explicit_counterexample`
Unit-cube rescaling (11)	`P1.unit_cube_H0_counterexample_family`
Arbitrarily small inputs	`P1.arbitrarily_close_unit_cube_counterexamples`
Proposition 5.3	`P1.same_family_H1_and_Euler_cancellation`
Theorem 1.3	`P3.p3_H0_presentation_counterexample_family`

Table 1: Public theorem-to-paper map.

All Lean names in table 1 share the root namespace `OneEdgeInstability`, which is omitted there.

The chain-level H_0 proof identifies the quotient by the span of oriented edge boundaries with the free vector space on connected components. For H_1, the proof identifies the range of the boundary with that same cellular span, applies rank-nullity, and constructs the two Hilbert measures before subtracting them. The transport lower bound follows directly from the marginal equations: every product-measure coupling is supported on the finite source-to-target product, where the disjoint integer lattices impose cost at least one per unit of mass. The upper bound is the vertical matching, written as a finite sum of Dirac measures on the product.

The checked source contains no `sorry`, `admit`, user-defined `axiom`, or `unsafe` declaration. For the audited public theorems, `#print axioms` reports only Lean's standard principles `propext`, `Classical.choice`, and `Quot.sound`.

The positive $n \leq 2$ half of corollary 1.2 is cited rather than reformalized. The minimality proof in proposition 3.1 and the subdivision argument in proposition 5.2 are paper proofs, not Lean theorems. Lean checks the counterexample family, the signed-measure and transport identifications, the rescaling, the module presentation theorem, and proposition 5.3.

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